Space-Time Π Identity and the Integrated Quadratic Boundary Oscillation (Route δ, Plan 2, Building Block SpaceTimeIdentity)

Abstract

This auxiliary block records the exact space-time radial-moment identity

$$\int_{\rho_*}^{\rho_\delta} \Pi(E_\rho, \bar{z}_\rho)\, d\mu(\rho) = (n-1)\int_{\widetilde{\Omega}} \sigma(x)\, dx,$$

and isolates the quantitative centroid-route estimate

$$\int_{\rho_*}^{\rho_\delta} \Pi(E_\rho, \bar{z}_\rho)\, d\mu(\rho) \le C_\Pi(n, R, \rho_*)\, \delta_T(\Omega). \tag{Π-int}$$

Conditional on (Π-int), we prove the integrated quadratic boundary oscillation

$$\int_{\rho_*}^{\rho_\delta}\int_{\partial^* E_\rho} |\nu_{E_\rho} - e_{r,\bar{z}_\rho}|^2\, d\mathcal{H}^{n-1}\, d\mu(\rho) \le C(n, R, \rho_*)\, \delta_T(\Omega), \tag{B^2-int}$$

and, under an additional annular/perimeter cap for the centroid slices, the integrated radial trace

$$\int_{\rho_*}^{\rho_\delta}\Big(\int_{\partial^* E_\rho} |H_{\bar{z}_\rho,\rho} - 1|\, d\mathcal{H}^{n-1}\Big)^2 d\mu(\rho) \le C(n, R, \rho_*)\, \delta_T(\Omega), \tag{T$_1^2$-int}$$

The proof of (T$_1^2$-int) uses a *slicing-then-squaring* decomposition $T_1 = T_1^{\mathrm{rad,slic}} + T_1^{\tan}$, in which the radial piece is bounded *linearly* (not at the FMP rate $\sqrt{\delta}$) via the good-ray equality of slicewise volumes and the bad-ray multiplicity bound of [1]; the squared volume Fraenkel $|E_\rho \Delta B_\rho|^2$ is controlled at rate δ via the new pointwise slice-rearrangement inequality $|E_\rho \Delta B_\rho|^2 \le C\, \Pi(E_\rho, \bar{z}_\rho)$. The repaired downstream metric route in [4] no longer depends on (Π-int), but this block remains a useful diagnostic for the centroid-based space–time approach. The invalid shortcut from the rearranged Talenti profile to (Π-int) has been removed; proving (Π-int) would be an additional structural theorem, not a prerequisite for the current metric closure.

1 Notation and statement of the targets

1.1 Standing hypotheses

Fix $n \ge 2$, $R \ge 2$, $\rho_* \in (0,1)$. Let $\Omega \subset B_R$ open with $|\Omega| = \omega_n$, $\delta := \delta_T(\Omega) \le \delta_0(n, R, \rho_*)$. Let $u = u_\Omega$ be the variational torsion function, $t : [\rho_*, 1] \to (0, \|u\|_\infty)$ the volume-radius parametrisation, $E_\rho := \{u > t(\rho)\}$, $t_B(\rho) := (1-\rho^2)/(2n)$, $\psi(\rho) := (-t_B'(\rho)) - (-t'(\rho)) = \rho/n + t'(\rho) \ge 0$, $\rho_\delta := 1 - \kappa\sqrt{\delta}$, $\mu(d\rho) := (-t'(\rho))d\rho$. By coarea applied to $u \in W_0^{1,2}$, E_ρ has finite perimeter for a.e. ρ.

For each ρ, $\bar{z}_\rho = \bar{z}_\rho$ is the centroid of E_ρ ([3] Theorem F). Since $E_\rho \subset B_R$, also $\bar{z}_\rho \in B_R$. No Cicalese–Leonardi containment is assumed at this stage; any lower bound on $|x - \bar{z}_\rho|$ must be justified separately on good levels or by truncating the kernel.

The L^2 divergence identity in [1]:

$$\int_{\partial^* E_\rho} |\nu_{E_\rho} - e_{r,\bar z_\rho}|^2 d\mathcal{H}^{n-1} = 2(P(E_\rho) - P(B_\rho)) + 2\Pi(E_\rho, \bar z_\rho), \tag{1}$$

where

$$\Pi(E_\rho, \bar z_\rho) := (n-1)\int_{B_\rho(\bar z_\rho)\setminus E_\rho} |x-\bar z_\rho|^{-1}dx - (n-1)\int_{E_\rho\setminus B_\rho(\bar z_\rho)} |x-\bar z_\rho|^{-1}dx \geq 0. \tag{2}$$

1.2 The three target theorems

Theorem 1.1 (Space-time Π identity, (G1))**.** $\int_{\rho_*}^{\rho_\delta} \Pi(E_\rho, \bar z_\rho)d\mu(\rho) = (n-1)\int_{\widetilde\Omega} \sigma(x)dx$ *for a signed Fubini density* σ *satisfying the kernel-weighted bound*

$$|\sigma(x)| \leq \int_{J_x} |x-\bar z_\rho|^{-1}\, d\mu(\rho),$$

where $J_x := \{\rho : x \in B_\rho(\bar z_\rho)\Delta E_\rho\}$.

Theorem 1.2 (Integrated (B^2-int) from (Π-int))**.** *Assume* (Π-int)*:*

$$\int_{\rho_*}^{\rho_\delta} \Pi(E_\rho, \bar z_\rho)\, d\mu(\rho) \leq C_\Pi(n, R, \rho_*)\, \delta_T(\Omega).$$

Then

$$\int_{\rho_*}^{\rho_\delta}\int_{\partial^* E_\rho} |\nu_{E_\rho} - e_{r,\bar z_\rho}|^2 d\mathcal{H}^{n-1}d\mu(\rho) \leq C_{B^2}(n, R, \rho_*)\, \delta_T(\Omega).$$

Theorem 1.3 (Auxiliary integrated radial trace, (T_1^2-int))**.** *Assume* (Π-int) *and the auxiliary annular/perimeter cap*

$\rho_*/2 \leq |x-\bar z_\rho| \leq 2R$ *on the relevant boundary pieces,* $\qquad P(E_\rho)-P(B_\rho)+\Pi(E_\rho, \bar z_\rho) \leq M_A(n, R, \rho_*)$

for a.e. $\rho \in [\rho_*, \rho_\delta]$*. Then*

$$\int_{\rho_*}^{\rho_\delta} \Big(\int_{\partial^* E_\rho} |H_{\bar z_\rho,\rho} - 1| d\mathcal{H}^{n-1}\Big)^2 d\mu(\rho) \leq C_{\mathrm{radial}}(n, R, \rho_*)\, \delta_T(\Omega).$$

2 Space-time identity (G1)

By (2), $\Pi(E_\rho, \bar z_\rho) = (n-1)\int_{\mathbb{R}^n}(\mathbf{1}_{B_\rho(\bar z_\rho)} - \mathbf{1}_{E_\rho})|x-\bar z_\rho|^{-1}dx$. By joint Borel measurability and Fubini,

$$\int_{\rho_*}^{\rho_\delta} \Pi(E_\rho, \bar z_\rho)d\mu(\rho) = (n-1)\int_{\mathbb{R}^n} \sigma(x)dx, \tag{3}$$

where $\sigma(x) := \int_{\rho_*}^{\rho_\delta}(\mathbf{1}_{B_\rho(\bar z_\rho)} - \mathbf{1}_{E_\rho})/|x-\bar z_\rho|\, d\mu(\rho)$. The integrand vanishes outside $\widetilde\Omega := B_{R+1}$ (which contains $\Omega \subset B_R$), because $\bar z_\rho \in B_R$ and $\rho \leq 1$. The kernel is locally integrable in dimension $n \geq 2$; if desired, one first proves the identity with $\min\{|x-\bar z_\rho|^{-1}, M\}$ and then lets $M \to \infty$ by monotone convergence applied to the positive and negative parts. Taking absolute values inside the ρ-integral gives the stated kernel-weighted crossing bound. $\square$

3 The remaining quantitative input

The identity of Theorem 1.1 rewrites $\int \Pi(E_\rho, \bar{z}_\rho)\, d\mu$ as an integral over the space–time mismatch

$$J_x := \{\rho \in [\rho_*, \rho_\delta] : x \in B_\rho(\bar{z}_\rho)\Delta E_\rho\}.$$

The rearranged Talenti/Nicola–Tilli profile gap controls the one-dimensional distributional profiles $t(\rho)$ and $t_B(\rho)$. It does *not*, by itself, control the physical crossing set J_x, because J_x also depends on the moving spatial radius $|x - \bar{z}_\rho|$. Therefore the implication

$$\mathcal{L}^1(J_x) \lesssim \delta_T(\Omega)$$

is not asserted here. The additional theorem that would finish this centroid subroute is the following integrated moment estimate.

Theorem 3.1 (Integrated Π-control; remaining centroid-route input)**.** *There is $C_\Pi(n, R, \rho_*) > 0$ such that, for every bounded open $\Omega \subset B_R$ with $|\Omega| = \omega_n$ in the small-deficit regime,*

$$\int_{\rho_*}^{\rho_\delta} \Pi(E_\rho, \bar{z}_\rho)\, d\mu(\rho) \le C_\Pi(n, R, \rho_*)\, \delta_T(\Omega). \tag{Π-int}$$

Remark 3.2 (Possible proof route)**.** A viable proof of Theorem 3.1 would have to combine the Fubini identity of Theorem 1.1 with a genuine space–time coarea or transport estimate for the physical crossings of $|x - \bar{z}_\rho| = \rho$. The H^1-in-ρ centroid estimate of [3] is the natural input for controlling the moving centre term, but an extra argument is still needed to convert the profile gap into the physical crossing-length bound.

4 Assembly of (B^2-int) from (Π-int)

Assume (Π-int). By (1) integrated,

$$\int |\nu - e_{r,\bar{z}_\rho}|^2 dH d\mu = 2\int (P - P_\rho) d\mu + 2\int \Pi(E_\rho, \bar{z}_\rho) d\mu.$$

The first piece is $\le C\delta$ via the deficit identity in [1]. The second piece is exactly (Π-int). Both pieces are $O(\delta)$, giving Theorem 1.2. □

5 The integrated radial trace (T$_1^2$-int) — slicing-then-squaring

Write $T_1 = T_1(E_\rho, \bar{z}_\rho) := \int_{\partial^* E_\rho} |\, |x - \bar{z}_\rho|/\rho - 1\, |\, d\mathcal{H}^{n-1}$ and decompose

$$T_1 = T_1^{\mathrm{rad,slic}} + T_1^{\mathrm{tan}},$$

with

$$T_1^{\mathrm{rad,slic}} := \int_{\partial^* E_\rho} ||x - \bar{z}_\rho|/\rho - 1| \cdot |e_{r,\bar{z}_\rho} \cdot \nu_{E_\rho}| d\mathcal{H}^{n-1},$$
$$T_1^{\mathrm{tan}} := \int_{\partial^* E_\rho} ||x - \bar{z}_\rho|/\rho - 1| \cdot (1 - |e_{r,\bar{z}_\rho} \cdot \nu_{E_\rho}|) d\mathcal{H}^{n-1}.$$

By the slicing identity [6, Thm 18.11] applied with $g(r) := |r/\rho - 1|$:

$$T_1^{\mathrm{rad,slic}} = \int_{S^{n-1}} \sum_{j=1}^{N(\theta)} |r_{\theta,j}/\rho - 1|\, r_{\theta,j}^{n-1} d\theta. \tag{4}$$

5.1 Pointwise bound for $T_1^{\text{rad,slic}}$

Lemma 5.1. $T_1^{\text{rad,slic}} \le C_4(n,R,\rho_*)[|E_\rho \Delta B_\rho(\bar z_\rho)| + (P(E_\rho) - P(B_\rho))]$.

Proof. Good rays. On a good ray ($N=1$), the slicewise 1D symmetric-difference *equals*:

$$\int_0^\infty |\mathbf{1}_{E_{\rho,\theta}} - \mathbf{1}_{(0,\rho)}| r^{n-1} dr = \frac{1}{n}|r_{\theta,1}^n - \rho^n|. \tag{5}$$

By MVT, $|r_{\theta,1}/\rho - 1| \cdot r_{\theta,1}^{n-1} \le \beta_n |r_{\theta,1}^n - \rho^n|$ with $\beta_n = \beta_n(R,\rho_*)$. Integrating and using spherical Fubini:

$$\int_{\{N=1\}} |r_{\theta,1}/\rho - 1| r_{\theta,1}^{n-1} d\theta \le n\beta_n |E_\rho \Delta B_\rho(\bar z_\rho)|.$$

Bad rays. $|r_{\theta,j}/\rho - 1| r_{\theta,j}^{n-1} \le C(R,\rho_*)$, summed gives $\le (2R)^{n-1}(1+k(\theta))$. We claim the bad-ray multiplicity bound

$$\int_{S^{n-1}} k(\theta)\, d\theta \;\le\; (\rho_*/2)^{-(n-1)} \,(P(E_\rho) - P(B_\rho)),$$

which we prove as follows (note: this estimate is conditional on the auxiliary annular cap of Theorem 1.3, which provides $|x - \bar z_\rho| \ge \rho_*/2$ on the relevant boundary pieces). For each direction $\theta \in S^{n-1}$, let $k(\theta) \ge 0$ count the crossings of the radial ray with $\partial^* E_\rho$ beyond the spherical baseline. By the radial-projection inequality and the area formula, $\int_{S^{n-1}} \sum_j r_{\theta,j}^{n-1}\, d\theta = P_{\text{rad}}(E_\rho) \le P(E_\rho)$, where $r_{\theta,j}$ are the crossing radii. Each crossing contributes $\ge (\rho_*/2)^{n-1}$ to this integral, since $|x - \bar z_\rho| \ge \rho_*/2$ on the annular cap. Summing the contributions beyond the spherical baseline gives $\int_{S^{n-1}} k(\theta)\, d\theta \le (\rho_*/2)^{-(n-1)}\,(P(E_\rho) - P(B_\rho))$, as claimed.

Combining the two estimates gives the claim. □

5.2 Pointwise bound for $T_1^{\tan}$

Lemma 5.2. $T_1^{\tan} \le (P(E_\rho) - P(B_\rho)) + \Pi(E_\rho, \bar z_\rho)$.

Proof. Under the auxiliary annular cap, $||x - \bar z_\rho|/\rho - 1| \le C(R,\rho_*)$, so $T_1^{\tan} \le C(R,\rho_*) \int (1 - |e_{r,\bar z_\rho} \cdot \nu|) dH \le C(R,\rho_*) \int (1 - e_{r,\bar z_\rho} \cdot \nu) dH = C(R,\rho_*)((P - P_\rho) + \Pi(E_\rho, \bar z_\rho))$ by the divergence identity in [1]. □

5.3 Squaring and integrating

By Lemmas 5.1, 5.2 and $(a+b+c)^2 \le 3(a^2+b^2+c^2)$:

$$T_1^2 \le 3C_5^2[|E_\rho \Delta B_\rho|^2 + (P - P_\rho)^2 + \Pi(E_\rho, \bar z_\rho)^2].$$

Integrating against $d\mu$ and bounding each piece by $C\delta$:

- $\int |E_\rho \Delta B_\rho|^2 d\mu \le C_8 \int \Pi(E_\rho, \bar z_\rho) d\mu \le C\delta$ via the slice-rearrangement (§6) + (G3).
- $\int (P - P_\rho)^2 d\mu \le M_A \int (P - P_\rho) d\mu \le C\delta$ under the auxiliary cap (uses $(P - P_\rho) \ge 0$ to bound $(P - P_\rho)^2 \le M_A(P - P_\rho)$ on the cap).
- $\int \Pi(E_\rho, \bar z_\rho)^2 d\mu \le M_A \int \Pi(E_\rho, \bar z_\rho) d\mu \le C\delta$ under the same cap.

Hence $\int T_1^2 d\mu \le C_{T_1^2}\delta$.

Combining with (G3) gives (T_1^2-int) as follows. Since $H_{\bar z_\rho,\rho} = (|x - \bar z_\rho|/\rho)\,(e_{r,\bar z_\rho} \cdot \nu)$, we decompose

$$H_{\bar z_\rho,\rho} - 1 \;=\; \big(|x - \bar z_\rho|/\rho - 1\big)\,(e_{r,\bar z_\rho} \cdot \nu) \;+\; \big(e_{r,\bar z_\rho} \cdot \nu - 1\big),$$

where on the annular cap $|e_{r,\bar z_\rho}\cdot\nu|\le 1$ and $\big||x-\bar z_\rho|/\rho-1\big|\le C(R,\rho_*)$, so both factors are bounded. Taking absolute values and integrating over $\partial^* E_\rho$ yields

$$\int_{\partial^* E_\rho}|H_{\bar z_\rho,\rho}-1|\,d\mathcal{H}^{n-1}\ \le\ T_1\ +\ \int_{\partial^* E_\rho}|e_{r,\bar z_\rho}\cdot\nu-1|\,d\mathcal{H}^{n-1}.$$

For the second term, using $|e_{r,\bar z_\rho}\cdot\nu-1|\le|e_{r,\bar z_\rho}-\nu|$ and applying Cauchy–Schwarz with respect to $d\mathcal{H}^{n-1}\,d\mu$ together with the perimeter bound $P(E_\rho)\le M_A$ on the bounded reduction,

$$\Big(\int|e_{r,\bar z_\rho}-\nu|\,d\mathcal{H}^{n-1}\,d\mu\Big)^2\ \le\ M_A\cdot\int|e_{r,\bar z_\rho}-\nu|^2\,d\mathcal{H}^{n-1}\,d\mu.$$

The L^2-bound on $|e_{r,\bar z_\rho}-\nu|^2$ provided by (G3) (Theorem 1.2) then gives an $O(\delta)$ control. Squaring the pointwise decomposition, applying $(a+b)^2\le 2(a^2+b^2)$, integrating against $d\mu$, and combining with $\int T_1^2\,d\mu\le C_{T_1^2}\delta$ yields (T_1^2-int). □

6 The slice-rearrangement inequality

Lemma 6.1. *Under the annular cap $\rho_*/2\le r\le 2R$, $|E_\rho\Delta B_\rho(\bar z_\rho)|^2\le C_8(n,R,\rho_*)\Pi(E_\rho,\bar z_\rho)$ pointwise in the smallness regime.*

Proof. Step 1: $\Pi(E_\rho,\bar z_\rho)\ge cM_1$. Let $M_1:=\int_{E_\rho\Delta B_\rho}|\rho-|x-\bar z_\rho||dx$. By rewriting (2),

$$\Pi(E_\rho,\bar z_\rho)=(n-1)\int_{E_\rho\Delta B_\rho}\frac{|\rho-|x-\bar z_\rho||}{\rho|x-\bar z_\rho|}dx\ge\frac{n-1}{2R\rho}M_1.$$

Step 2: $|E_\rho\Delta B_\rho|^2\le C\,M_1$. By polar coordinates and a slicewise Cauchy–Schwarz on $[\rho_*/2,2R]$, $\sigma(\theta)^2\le 4Rm_1(\theta)$ with $\sigma(\theta)=\mathcal{L}^1(E_{\rho,\theta}\Delta(0,\rho))$, $m_1(\theta)=\int|r-\rho|\mathbf{1}_{E_{\rho,\theta}\Delta(0,\rho)}dr$. Cauchy–Schwarz on S^{n-1} then gives $(\int\sigma d\theta)^2\le 4R|S^{n-1}|\int m_1 d\theta$. Since $\sigma(\theta)\,d\theta\cdot r^{n-1}$ (for $\rho_*/2\le r\le 2R$) gives the volume element up to the comparable factors $(\rho_*/2)^{n-1}$ and $(2R)^{n-1}$, the slice-to-volume conversion preserves the bound up to constants depending only on n,ρ_*,R; this yields $|E_\rho\Delta B_\rho|^2\le C_9\,M_1$.

Combining Steps 1–2 gives the claim. □

7 Discussion: why T_1-route closes where T_2-route stalls

Remark 7.1. The T_2-route would extract on good rays $(r_{\theta,1}/\rho-1)^2 r_{\theta,1}^{n-1}\le C|r_{\theta,1}^{n-1}-\rho^{n-1}|$ (a square on LHS, unsigned mismatch on RHS), then need $\int|r_{\theta,1}^{n-1}-\rho^{n-1}|d\theta\le C\sqrt{\delta_T}$ (FMP rate). Integrating against $d\mu$ leaks at $\sqrt{\delta}$.

The T_1-route uses linear good-ray equality (5) and the slice-rearrangement to get $|E_\rho\Delta B_\rho|^2\le C\Pi$, which integrates to $C\delta$ via (G3). Squaring *before* integrating avoids the FMP bottleneck.

8 Conditionality

Inputs used after (Π-int) and the auxiliary annular/perimeter cap: slicing identity (Maggi); bad-ray multiplicity bound (elementary, see proof of Lemma 5.1); slicewise good-ray equality; spherical Fubini; divergence identity (5.3); Talenti profile gap; slice-rearrangement Lemma 6.1; (F) of [3].

Inputs explicitly not used: the per-ρ pointwise radial-trace inputs (1.1), (1.2)=(R), (2.4); pointwise (B^2); the boundary-layer pointwise (7.1) of [5]; any selection, graph entry, Schauder, Fuglede.

Centroid-route status: Theorem 3.1, plus the annular/perimeter cap stated in Theorem 1.3, would complete this centroid subroute. Without them the present block proves the exact space–time identity and a conditional centroid closure only. The repaired metric closure is instead supplied by [4].

References

[1] Plan 2 building block LevelSetIdentity.

[2] Plan 2 building block MetricFramework.

[3] Plan 2 building block CentroidBound.

[4] Plan 2 building block WeightedMetricTrace.

[5] Plan 2 building block BoundaryLayerTransfer.

[6] F. Maggi, Cambridge Univ. Press, 2012. Thm 15.9, 18.11, 19.4.